

Parametric 3D CAD Model of a Steel Girder Bridge

Technical Implementation Report

Omkar Kottawar

February 3, 2026

Contents

1	Summary of the Implementation	2
2	Project Overview	2
2.1	Problem Statement	2
2.2	Scope of the Model	2
3	System Architecture and Design	2
3.1	Object-Oriented Design	2
3.2	Design Principles	2
4	Geometric Modeling Framework	3
4.1	Input Parameters	3
4.2	Coordinate System	3
4.3	Derived Geometric Calculations	3
5	Implementation Details	4
5.1	Component Geometry Creation	4
5.2	Geometric Transformations	4
5.3	Assembly Process	4
6	Visualization and Export	4
7	Testing and Validation	4
8	Conclusion	4

1 Summary of the Implementation

This report documents the implementation of a parametric 3D CAD model of a short-span steel girder bridge superstructure developed using the `pythonOCC` library. The objective of the project is to create a modular and reusable modeling framework in which bridge geometry is controlled through adjustable parameters. The system generates a complete three-dimensional representation of the bridge, supports visualization within the Open-CASCADE environment, and provides export functionality to standard CAD formats such as STEP and BREP. The implementation emphasizes parametric design principles, object-oriented architecture, and geometric transformation techniques.

2 Project Overview

2.1 Problem Statement

The task involves developing a script capable of generating a parametric 3D model of a steel girder bridge superstructure. The model must be assembled using component-based factory functions and allow all primary geometric dimensions to be modified through parameters. The final solution should support visualization, modularity, and export of the generated CAD geometry.

2.2 Scope of the Model

The model focuses exclusively on the bridge superstructure and includes:

- Longitudinal steel I-section girders
- Concrete deck slab
- Parapets along deck edges

Substructure components such as piers, bearings, and foundations are outside the scope of this implementation.

3 System Architecture and Design

3.1 Object-Oriented Design

The implementation follows an object-oriented approach to ensure modularity and separation of responsibilities. Each physical component of the bridge is represented as a class that manages its geometry, while common CAD operations such as translation and rotation are handled by a base class.

3.2 Design Principles

The following design concepts are applied:

- **Parametric Modeling:** All geometric dimensions are defined by variables.

Class	Role	Key Responsibilities
BridgeComponent	Base class for CAD entities	Shape storage, translation, rotation
Girder	Steel I-section member	Geometry creation using section parameters
Deck	Concrete slab	Rectangular prism generation
Parapet	Safety barrier	Prism generation and placement
BridgeModel	Assembly controller	Layout computation, component creation, assembly, visualization

- **Modularity:** Components are created independently and assembled later.
- **Separation of Concerns:** Geometry definition and spatial positioning are handled separately.
- **Reusability:** Classes can be extended for additional bridge components.

4 Geometric Modeling Framework

4.1 Input Parameters

All primary bridge dimensions are defined as parameters at the beginning of the script. These include span length, number of girders, girder spacing, section dimensions, deck thickness, parapet dimensions, and optional skew angle. This structure allows the same script to generate multiple bridge configurations.

4.2 Coordinate System

A right-handed coordinate system is used:

- X-axis: Bridge longitudinal direction
- Y-axis: Transverse direction
- Z-axis: Vertical direction

All components are initially created at the origin and subsequently positioned using geometric transformations.

4.3 Derived Geometric Calculations

Several geometric properties are calculated from input parameters:

$$\text{Deck Width} = \text{Spacing} \times (N_{\text{girders}} - 1) + 2 \times \text{Overhang}$$

Girder positions are distributed symmetrically about the global Y-axis to maintain geometric balance. The deck elevation is set equal to the girder depth so that it rests on the girders.

5 Implementation Details

5.1 Component Geometry Creation

Component solids are generated using factory functions. The girder is formed as an I-section by combining rectangular solids representing flanges and web. The deck and parapets are modeled as rectangular prisms.

5.2 Geometric Transformations

All positioning operations are performed using OpenCASCADE transformation classes. Translation vectors move components to their computed coordinates, while optional rotation is applied to introduce bridge skew.

5.3 Assembly Process

Individual component shapes are combined into a compound topology representing the full bridge model. This compound is used for visualization and export.

6 Visualization and Export

The model is displayed in an interactive OpenCASCADE window with appropriate view fitting and component coloring. Export functionality enables saving the assembled model to STEP or BREP formats for compatibility with external CAD software.

7 Testing and Validation

Multiple bridge configurations were generated to verify parameter handling, transformation accuracy, and component placement. Tests confirmed that changes in span length, number of girders, spacing, and skew angle produce correct geometric updates.

8 Conclusion

The implemented system demonstrates a structured approach to parametric CAD modeling using pythonOCC. The object-oriented architecture, combined with transformation-based assembly, provides a flexible framework capable of generating various steel girder bridge configurations. The design supports extensibility for future structural components or modeling enhancements.